

REGINA ADOBEA ESSIEN

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PROFESSIONAL SUMMARY

Data Scientist with over 3+ years of experience applying statistical analysis, data mining, machine learning, and AI-powered analytics to solve business and operational problems. Skilled in data cleaning, data validation, feature engineering, predictive modeling, dashboard development, KPI tracking, and stakeholder reporting, with hands-on experience across Power BI, PostgreSQL, Spark, BigQuery, TensorFlow, PyTorch, and AWS. Experienced in building data pipelines, machine learning models, and natural-language analytics tools for dealer performance, loan approval, fraud detection, churn prediction, and customer analytics. Strong ability to translate complex datasets into clear insights, recommendations, and decision-support tools for technical and non-technical stakeholders.

TECHNICAL SKILLS

- **Programming and Databases:** Python, SQL, PostgreSQL, MySQL, BigQuery, Excel, SPSS
- **Data Science & ML:** Pandas, NumPy, Scikit-learn, XGBoost, SHAP, Predictive Modeling, Regression, Classification, Time-Series Forecasting, Market Forecasting, Feature Engineering, Cross Validation, Model Evaluation
- **AI & Deep Learning:** LangChain, Google ADK, RAG Pipelines, AI Agents, LLM-Powered Analytics, Prompt Engineering, Hugging Face, TensorFlow, PyTorch, Keras, CNNs
- **Big Data & Cloud:** Spark, PySpark, Spark MLlib, Hadoop, Databricks, AWS
- **Data Engineering & Workflow:** ETL Pipelines, Data Pipelines, Data Ingestion, Data Cleaning, Feature Generation, Workflow Automation, API Integration, Model Deployment, Docker, CI/CD, Git, GitHub, Technical Documentation
- **Visualization & BI:** Power BI, Excel Dashboards, Data Visualization, KPI Reporting, Stakeholder Reporting

WORK EXPERIENCE

Dealer Engagement Analyst, Autochek Ghana, Accra (Jan 2024- Present)

- Developed machine learning models and analytical dashboards using dealer and customer data to predict financing product suitability, including Equity Release, Dealer Financing, and Import Financing, supporting targeted sales decisions across 2,000+ car dealers and improving reporting efficiency by 20%.
- Designed dashboards and technical business intelligence solutions using SQL and Power BI, reducing manual reporting effort by 20%.
- Built a Dealer Copilot to support analytics, reporting, and business intelligence, reducing report generation time by 25% and improving access to insights for business teams.
- Developed an AI-powered loan intake and sourcing assistant using TypeScript and Lua to collect applicant details, qualify leads, and structure loan application data for follow-up, improving consistency across loan sourcing workflows.
- Supported data architecture and data flow across analytics systems, internal platforms, and reporting workflows, improving data reliability and reducing inconsistencies by 15%.

Data Analyst, Dimension Data, Accra (Apr 2023-Jan 2024)

- Cleaned, processed, and analysed contract and sales datasets using Python, SQL, and Excel, improving data accuracy by 80%.
- Built interactive Power BI dashboards and reporting solutions to monitor customer behaviour, contract performance, and sales trends, contributing to a 20% increase in sales productivity.
- Performed data analysis to identify churn patterns, customer trends, and operational insights used to support business decisions.
- Automated recurring reporting workflows using Excel macros and SQL, reducing report generation time by 40%.

Data Analyst/Data Scientist Trainee, 131 Sunflower Hub, Accra (Jun 2022 -Dec 2022)

- Built a predictive model to support fraud monitoring in banking and mobile money transaction datasets using statistical and machine learning techniques.
- Collected, processed, and analysed structured and web-scraped data using Python to support data science and analytics use cases
- Applied statistical modelling to forecast gas prices and uncover market trends from historical data.
- Designed and built a database for grocery pricing data from 5 supermarkets, supporting efficient data

- storage, retrieval, and analysis.
- Developed dashboards and visual reports to analyse employee turnover and generate actionable business insights.

Data Entry/Research Assistant, Stratcomm Africa, Accra (Aug 2020- Aug 2021)

- Conducted sentiment analysis and trend tracking: Analyzed news data from newspapers and TV before the 2020 elections using SPSS, providing data-driven insights that influenced campaign strategies.
- Processed and structured large datasets in SPSS to generate insights, contributing to a strategy that secured an additional 10,000 votes in a key region.
- Developed client reports and visualized key trends by translating data findings into clear reports that guided strategic decision-making for political campaigns.
- Automated data summarization using Excel and SPSS to streamline the data entry process, ensuring real-time updates and reducing manual processing time.

Data & Customer Insights Intern, Ecobank Ghana, Accra (Jun - Aug 2018)

- Analyzed customer data by processing and validating data on 100+ customer accounts, ensuring accuracy in onboarding and account creation.
- Performed data-driven customer segmentation by analyzing patterns in account type inquiries and investment product interest, helping improve targeted service delivery.
- Prepared daily analytical reports by compiling summary reports on customer interactions and service trends, providing insights that enhanced branch operations.
- Automated report generation using Excel functions to streamline daily reports, reducing manual work and improving efficiency.

PROJECTS

- Developed a loan approval prediction model using applicant and financial features, applying data cleaning, feature engineering, model evaluation, and SHAP explainability to support risk analytics and data-driven lending decisions.
- Built and deployed a binary income prediction model using PySpark MLlib, applying feature assembly, logistic regression, hyperparameter tuning, 5-fold cross-validation, and evaluation with AUC-ROC, precision, recall, and confusion matrix analysis.
- Built and evaluated a customer model to identify customer churn patterns using Python, feature engineering, and predictive modelling techniques to support retention-focused insights.
- Built and deployed agentic AI analytics copilot using RAG, SQL-backed tools, and task-specific agent workflows to help business users query dealer performance, sales, leads, applications, and operational metrics in natural language.
- Built a deep learning melanoma image classification model using 13,900 skin lesion images, applying TensorFlow, Keras, CNNs, transfer learning models including ResNet50, EfficientNet, DenseNet, and MedFormer, AUC-ROC evaluation, confusion matrix analysis, and Grad-CAM interpretability to distinguish benign and malignant cases.

EDUCATION

- **M.S.c Data Science and Analytics** - Academic City University, Accra
- **B.A Political Science and English** – University of Ghana, Legon, Accra

LEADERSHIP, AWARDS & VOLUNTEERING

- **Best Computational Science and Informatics Student Award** -Recognized for outstanding academic performance and contribution in computational science and informatics.
- **Communications Committee & Tutorial Facilitator (Ghana Data Science Community / Women in Machine Learning and Data Science)** Facilitated beginner-friendly SQL and data science sessions for women, covering database creation, data management, SQL queries, ordering, grouping, and aggregation.

CERTIFICATIONS

- Certified AWS AI Practitioner
- AWS AI Practitioner Challenge (Udacity)
- Kaggle Introduction to and Intermediate Machine Learning
- Google Data analytics Professional Certificate